

An investigation of hydrogen depth profiling using ToF-SIMS

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Hydrogen depth distributions in silicon, zinc oxide, and glass are of great interest in material research and industry. Time-of-flight SIMS has been used for hydrogen depth profiling for many years. However, some critical information, such as optimal instrumental settings and detection limits, is not easily available from previous publications. In this work, optimal instrumental settings and detection limits of hydrogen in silicon, zinc oxide, and common glass were investigated. The recommended experimental settings for hydrogen depth profiling using time-of-flight SIMS are: (i) keeping pressure in the analysis chamber as low as possible, (ii) using a cesium beam for sputtering and monitoring the H^- signal, (iii) employing monatomic ion analysis beams with the highest currents, and (iv) using interlace mode. In addition, monatomic secondary ions from a matrix are recommended as references to normalize the H^- signal. Detection limits of hydrogen are limited by the pressure of residual gases in the analysis chamber. The base pressure of the analysis chamber (with samples) is about 7×10^{-10} mbar in this study, and the corresponding detection limits of hydrogen in silicon, zinc oxide, and common glass are 1.3×10^{18} atoms/cm³, 1.8×10^{18} atoms/cm³, and 5.6×10^{18} atoms/cm³, respectively. Copyright © 2011 John Wiley & Sons, Ltd.

Keywords: hydrogen depth profiling; ToF-SIMS; detection limit; optimal instrumental settings

Introduction

Hydrogen depth profiling is of great importance in material research and industry applications. For example, it has been reported that low concentration of hydrogen can greatly affect the optical and electric performance of some semiconductor materials, such as zinc oxide^[1–4] and silicon.^[5–8] In addition, water concentration in glass is of great interest in the vacuum glass industry and in glass corrosion research.^[9–12] There is no technique available to directly determine water depth distribution in glass, and a common alternative method requires using hydrogen depth profiling with an assumption that all hydrogen comes from water molecules.

Only a few techniques can be used to track the depth distribution of hydrogen in these materials. The three major techniques include SIMS, nuclear reaction analysis (NRA) and elastic recoil detection analysis (ERDA).^[13–17] Among these techniques, SIMS provides the best sensitivity and depth resolution. In SIMS depth profiling experiments, a cesium beam typically is used for sputtering, and the H^- ion is monitored because H^- is more sensitive than H^+ .^[18]

Generally, three types of commercial SIMS instruments are available based on their mass analyzers. They are magnetic SIMS, quadrupole SIMS, and time-of-flight SIMS (ToF-SIMS). Hydrogen depth profiling in silicon using quadrupole SIMS was reported over 30 years ago.^[14,15] The detection limit of hydrogen in silicon was found to be about $\leq 2 \times 10^{18}$ atoms/cm³, and it was limited by hydrogen-contained residual gases, mostly water and hydrogen gas, in the analysis chamber.

Usually, ToF-SIMS is used for surface analysis, but it can also be used for depth profiling tasks. The currents of analysis ion beams of ToF-SIMS instruments are quite small (pA level) and a dual-beam strategy is usually used to perform depth profiling tasks, where a sputtering ion beam with a high current is introduced to sputter materials away at a reasonable rate so that depth

profiling measurements can be done in a comfortable time range.^[19] Because most materials are sputtered away by the sputtering beam – not the analysis beam – the sensitivity of ToF-SIMS is about 1–2 orders of magnitude poorer than that of magnetic or quadrupole SIMS. Nonetheless, it is not difficult for ToF-SIMS to reach a detection level of 1×10^{18} atoms/cm³ for hydrogen depth profiling.^[3,4,20] Therefore, ToF-SIMS has been widely used in hydrogen depth profiling.

Although hydrogen depth profiling using ToF-SIMS has been reported in many publications, some critical information, such as the optimal experimental settings and detection limits, is not easy accessible. Therefore, it is inconvenient for common ToF-SIMS users to find a guide for their hydrogen depth profiling work. Consider the following:

- Cluster ion analysis beams are available for new commercial ToF-SIMS instruments. However, are they good choices for hydrogen depth profiling?
- Two analysis modes are regularly used in dual-beam ToF-SIMS depth profiling. One is called 'interlace' mode with quasi-parallel sputtering to analysis, and the other is called 'noninterlace' mode, where analysis cycles are separated from sputter cycles.^[19] Is there any difference between them for hydrogen depth profiling?
- More importantly, what are the detection limits for hydrogen depth profiling in some common materials (e.g. silicon, zinc oxide, and glass)?

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- In addition, a common strategy in quantifying the concentration of hydrogen with SIMS is using ion-implanted reference samples. It is a well-known ToF-SIMS advantage that all secondary ion species can be collected and data reconstruction is feasible. If so, both monatomic secondary ions and cluster secondary ions can be used as references for normalization. Still, a question remains: is there any difference between them?

In this work, hydrogen depth profiling of silicon, zinc oxide, and common glass samples were studied. Optional experimental settings, practical detection limits, and the best data normalization strategy were investigated.

Experimental section

Sample preparation

Deuterium in silicon (D/Si), deuterium in zinc oxide (D/ZnO), and deuterium in glass (D/glass) samples were prepared at the Environmental Molecular Sciences Laboratory located at Pacific Northwest National Laboratory (PNNL). In brief, 300 keV D^+ was implanted into a silicon (100) wafer (Silicon Inc.), a high-quality pure zinc oxide crystal (CrysTech Inc.), and a common glass sample (Fisher Scientific Inc. premium microscope slide [plain]) with an incident angle of 60 degrees. All samples were cut to a size of about $1 \times 1 \text{ cm}^2$. The total D^+ dose was about 1×10^{15} atoms/ cm^2 . In addition, two hydrogen-implanted samples, an H/Si sample^[21] and an H/ZnO sample,^[22] also were studied. The H/Si sample was prepared by implanting 6 keV H^+ ions into a silicon substrate with a total H^+ dose of 5×10^{16} atoms/ cm^2 . The H/ZnO sample was prepared by implanting 50 keV H^+ ions into a zinc oxide crystal with a total H^+ dose of 1×10^{15} atoms/ cm^2 .

Time-of-flight SIMS

Time-of-flight SIMS measurements were performed using a TOF-SIMS 5 spectrometer (IONTOF GmbH, Münster, Germany). It was equipped with a 25 keV bismuth (Bi) cluster ion source, a 10 keV C_{60}^+ source, and a 0.2–2.0 keV Cs^+/O_2^+ sputtering source.

The sputtering ion beam was a 2.0 keV Cs^+ beam, which was rastered over a $300 \times 300 \mu\text{m}^2$ area with an incident angle of 45 degrees during sputtering. The sputtering beam's current was between 120–150 nA range, which varied day-to-day, but it was relatively stable in each measurement (typically <1% variation). Cesium sputtering was used, and the H^- ions were detected because H^- signal is more sensitive than H^+ signal.^[18] A pulsed 25 keV Bi^+ beam was used for analysis. The pulsed Bi^+ analysis beam was rastered over a $100 \times 100 \mu\text{m}^2$ area at the center of Cs^+ sputter craters during data acquisition. The pulse frequency of the Bi^+ beam was 10 kHz and the corresponding current was about 1.0 pA. The base pressure of the analysis chamber was about 5×10^{-10} mbar. The samples were pumped down in the introduction chamber overnight before they were introduced into the analysis chamber, so that the pressure in the analysis chamber could be kept at about 7×10^{-10} mbar with samples in it. Residual gas analysis showed that the components of residual gases were 70% H_2 , 15% H_2O , and 15% N_2 and CO_2 . Silicon and zinc oxide were semiconductors and charge compensation was not used for them. During the silicon and zinc oxide depth profiling experiments, the pressure in the analysis chamber increased a little and it was about 1.1×10^{-9} mbar. However, charge compensation must be used for the glass samples, and the

corresponding pressure increased to about 3.0×10^{-9} mbar. After SIMS measurements, the samples were removed from the vacuum, and the depths of sputter craters were measured using a stylus profilometer (Dektak 150, Veeco Instruments, Inc.). A constant sputtering rate was assumed for each depth profiling measurement.

Sample annealing

A common glass sample contains a large amount of water in its surface layer and trace water in bulk. To determine the detection limit of hydrogen in glass samples, this water must be removed. A thin glass sample (0.13-mm-thick cover-slips, Ted Pella Inc.) was analyzed before and after annealing. Two annealing temperatures were performed (600 and 950 °C), and the annealing time was 2 h. These two temperatures were used because the glass transition temperature of common glass is about 560 °C, and the liquidus temperature is about 1000 °C. The annealing was performed in a vacuum chamber with a pressure lower than 1.0×10^{-5} mbar.

Results and discussions

It has been reported that the detection limits of hydrogen depth profiling are limited by residual gases in the vacuum system.^[14,15] To assess the effect of residual gases, three different pressures were tested, and the results are shown in Figure 1. The results indicate that the pressure in the analysis chamber dramatically affects the detection limits of hydrogen. The H^- signal intensity at 1.4×10^{-7} mbar is about 40 times the H^- signal intensity at 4.0×10^{-9} mbar, or is about 160 times the H^- signal intensity at 1.1×10^{-9} mbar. Apparently, a key factor to obtain the best detection limit is to keep the pressure as low as possible. However, every commercial ToF-SIMS instrument has its own vacuum limit for its analysis chamber. A common value is 5×10^{-10} mbar (or Torr). If better vacuum is needed, extra efforts, either using more powerful pumps or using long time baking, are required. This is not practical for routine sample analysis. In our study, $4\text{--}5 \times 10^{-10}$ mbar base pressure in the analysis chamber was routinely available after weekend baking. This pressure increased a little to $\sim 7 \times 10^{-10}$ mbar after a clean silicon wafer was introduced in it if the sample was overnight pumped in the

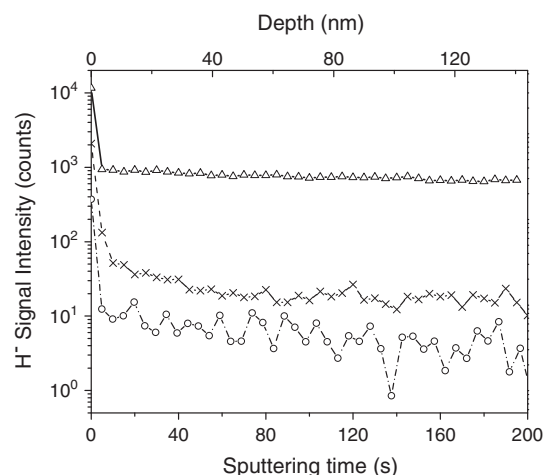


Figure 1. Hydrogen depth profiles of a clean silicon (100) wafer at three different pressures. Triangles: 1.4×10^{-7} mbar; crosses: 4.0×10^{-9} mbar; circles: 1.1×10^{-9} mbar.

introduction chamber. During the sputtering, the pressure in the analysis chamber increased to $\sim 1.1 \times 10^{-9}$ mbar. For routine measurements, this is the best obtainable vacuum.

Usually, Cs^+ sputtering is used for hydrogen depth profiling because H^- is more sensitive than H^+ ,^[18] and Cs^+ sputtering can enhance yields of negative ions.^[23] In our instrument, C_{60}^{n+} ($n = 1-3$) sputtering is also available. It is found that the D^- signal in a zinc oxide sample with C_{60}^{2+} sputtering is only $\sim 20\%$ the D^- signal with Cs^+ sputtering (Fig. 2). As such, Cs^+ sputtering is critical for hydrogen depth profiling. Normally, O_2^+ sputtering is not recommended for hydrogen depth profiling, because the O_2^+ source introduces high pressure ($\sim 5 \times 10^{-8}$ mbar) into the analysis chamber. This high pressure does lead to a high background H^+ signal, even if a high purity O_2 gas is used.

Cluster primary ions have been introduced for ToF-SIMS analysis for more than 10 years, and they provide 10–1000 times signal enhancement for molecular ions when compared to monatomic primary ions.^[24,25] Our instrument can provide five cluster ion analysis beams (Bi_3^+ , Bi_3^{2+} , Bi_5^+ , C_{60}^+ , and C_{60}^{2+}) for practical analysis. However, our previous results show that these cluster primary ions only provide limited enhancement for H^- signals, and the Bi^+ beam still provides the highest H^- signal because it offers much stronger currents than those cluster ion analysis beams.^[26] It is worth noting that, for most commercial ToF-SIMS instruments, monatomic ion analysis beams from liquid metal ion guns provide much stronger currents than any cluster ion analysis beams. Therefore, the monatomic ion analysis beam with the highest beam current is recommended for hydrogen depth profiling work.^[26]

Two modes are usually used for dual-beam depth profiling. One is 'interlace' mode, where the sputtering beam is active between pulses of the analysis ion beam. The other is called 'noninterlace' mode, where analysis time zones (up to several seconds each) are separated from sputtering time zones (up to several seconds each), and relaxation time zones (up to several seconds each) are added between them to reduce the charging effect on the sample surface.^[19] For conductive or semiconductive samples, the interlace mode depth profiling is preferred because it usually takes less time and it can keep entire depth information, even for a very thin layer. In our experiments, the interlace mode

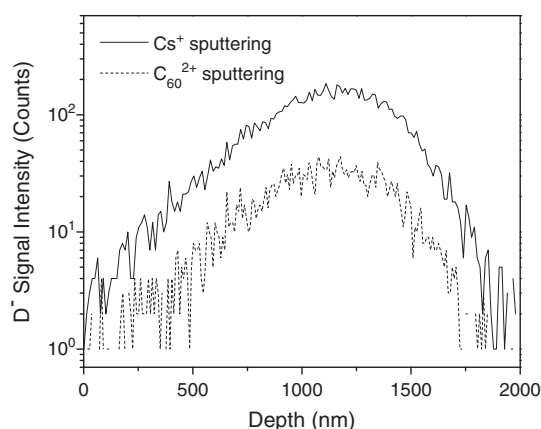


Figure 2. A comparison of D^- signal intensity in a deuterium-implanted zinc oxide sample using 2 keV Cs^+ and 20 keV C_{60}^{2+} as sputtering ions. The analysis ion beam is a pulsed 25 keV Bi^+ beam with a 1.0 pA current at a 10 kHz repeating rate, and the collecting time for each data point is about 16.38 s.

performs very well for the silicon and zinc oxide samples. Figure 3 shows an example of silicon depth profiling. It is shown that H^- and O^- background signals are very low when interlace mode is used. However, if the Cs^+ sputtering beam stops, H^- and O^- signals dramatically increase in 5 s (>10 times), and they continuously increase with time (Fig. 3). The increasing H^- and O^- signals can be explained by the adsorption of residual water and hydrogen molecules on the sample surface. Apparently, if the noninterlace mode is used for hydrogen depth profiling, a very strong background H^- signal appears, which ruins the detection limit.

Normally, depth profiling glass samples is partial to using the noninterlace mode because charge compensation is easy and effective using this mode. However, this is not a good choice for hydrogen depth profiling. To achieve a decent detection limit, the interlace mode must be used. Fortunately, the interlace mode is practical for glass depth profiling if a suitable charge compensation voltage (reflector voltage) is used. For example, 350 V was used in this work and the results are quite acceptable. It should be noted that this value may be a case-dependent value, and may change sample-to-sample or instrument-to-instrument.

Secondary ion signal intensities may change because of sample topography, sample locations, and day-to-day variations in instrumental parameters, especially ion beam currents. For example, H^- signals at different locations on a glass sample are shown in Fig. 4(a). The H^- signal from the sample corner is significantly lower than the H^- signals from the sample center. Therefore, data normalization is required for quantification. Generally, a signal coming from a matrix with medium intensity can be used as a reference, because a weak signal may have large variation, while a strong signal may saturate the detector. In addition, Poisson correction was used to correct all data in this study. One advantage of ToF-SIMS is that all secondary ion species can be collected during experiment, and data reconstruction is feasible. Thus, many species can be chosen as references during data analysis. For example, $^{18}\text{O}^-$, $^{28}\text{Si}^-$, $^{30}\text{Si}^-$, SiO_2^- and SiO_3^- show reasonable signal intensities during glass depth profiling. Figures 4(b), (c), and (d) show normalization results of H^- , $^{18}\text{O}^-$, and SiO_2^- (respectively) using $^{28}\text{Si}^-$ as a reference. After normalization, the H^- and $^{18}\text{O}^-$ signals show quite consistent

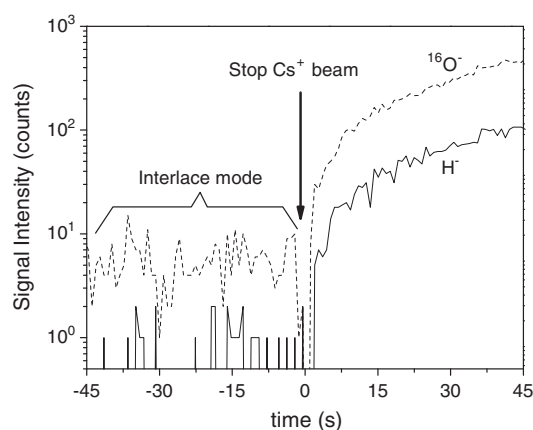


Figure 3. A depth profiling comparison of interlace mode to noninterlace mode of a clean silicon (100) wafer. Solid line, H^- signal; dashed line, $^{16}\text{O}^-$ signal. The primary ion beam is a pulsed 25 keV Bi^+ beam with a 1.0 pA current at a 10 kHz repeating rate, and the collecting time for each data point is about 0.80 s. Before the 0 s point, interlace mode is used. After the 0 s point, the Cs^+ sputtering beam is stopped but the Bi^+ primary ion beam is still active for measurement. The pressure in the analysis chamber is $\sim 1.1 \times 10^{-9}$ mbar during sputtering.

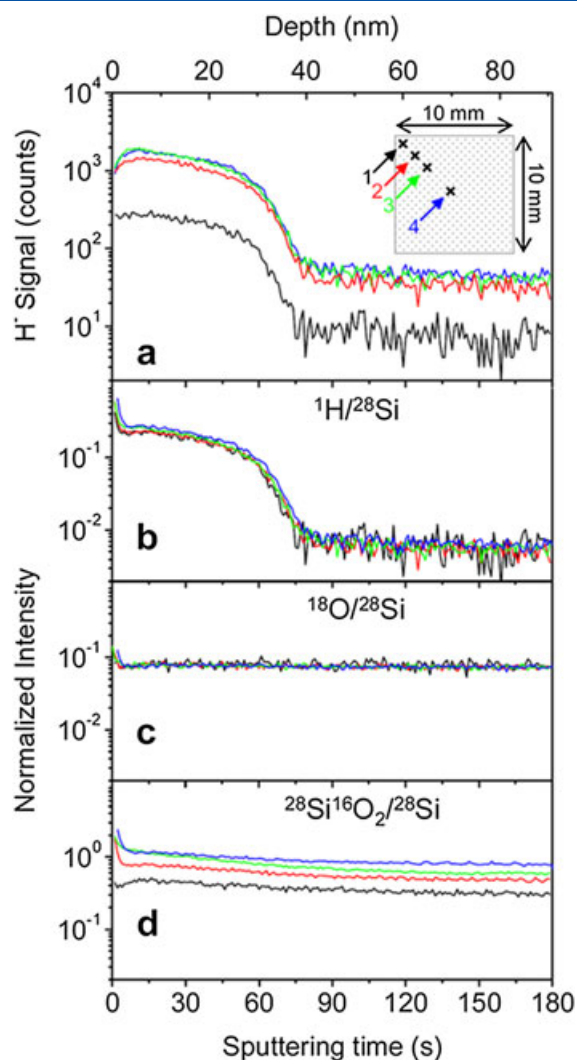


Figure 4. A comparison of data normalization of depth profiles of a Fisher brand glass sample. (a) H^- signal depth profiles from different locations (as shown in the insertion) on a $\sim 10 \times 10 \text{ mm}^2$ glass sample; (b) H^- signals normalized by $^{28}\text{Si}^-$ signals; (c) $^{18}\text{O}^-$ signals normalized by $^{28}\text{Si}^-$ signals; (d) $^{28}\text{Si}^{16}\text{O}_2^-$ signals normalized by $^{28}\text{Si}^-$ signals. Black, 1# location (0.5 mm, 0.5 mm from top left); red, 2# location (1.5 mm, 1.5 mm); green, 3# location (2.5 mm, 2.5 mm); blue, 4# location (4.5 mm, 4.5 mm).

values among all positions, but the SiO_2^- signals vary greatly. For normalizing H^- signals, it seems that monatomic secondary ion signals are more suitable to be the reference than cluster secondary ion signals. A possible explanation for this observation is that monatomic secondary ions may share similar energy and angular distributions, which may differ from those of cluster secondary ion species.^[27] Therefore, the collection efficiencies of monatomic

secondary ion species in an electric field may be relatively uniform, and it may be different from the collection efficiencies of cluster secondary ion species.

Relative sensitivity factors (RSFs) have been widely used in elemental quantification of SIMS depth profiling. RSFs of D and H in this study are listed in Table 1. Our RSF values of D and H in silicon ($\text{D} = 1.8 \times 10^{23}$ and $\text{H} = 1.4 \times 10^{23}$) are considerably lower than the document values ($\text{D} = 4.3 \times 10^{23}$ and $\text{H} = 3.0 \times 10^{23}$).^[18] It seems RSF values may vary instrument-to-instrument, and the document values from traditional magnetic or quadrupole SIMS may not be directly used in ToF-SIMS dual beam depth profiling.

Previous documentation shows that H^- is a little more sensitive ($\sim 30\text{--}40\%$) than D^- in silicon, gallium arsenide, and diamond matrices.^[18] In this study, two H-implanted reference samples (an H/Si sample^[21] and an H/ZnO sample^[22]) also were analyzed. The results show that H is indeed a little more sensitive than D (Table 1). The sensitivity difference between H and D is about 20–30%. Combining our results and document values, if D is used as a reference to quantify H concentrations in a matrix, about 30% more sensitivity (H over D) may be considered.

Detection limits may be one of the most important parameters for normal ToF-SIMS users. To quantify H^- background signals, D-implanted samples with known doses were analyzed, and the D^- signals were used to quantify the H^- signals. Figure 5 (top) shows that the detection limit of H in silicon is about 1.8×10^{18} atoms/ cm^3 (1.3×10^{18} if the H/D sensitivity difference is considered). This value is in agreement with the previously reported value using quadrupole SIMS.^[15] Figure 5 (bottom) shows that the detection limit of H in zinc oxide is about 2.5×10^{18} atoms/ cm^3 (1.8×10^{18} if the H/D sensitivity difference is considered).

Figure 6 (top) shows H^- and D^- depth profiles of the D/glass sample. It is shown that the H^- background is about 1.2×10^{20} atoms/ cm^3 . This value is significantly higher than those recorded in silicon or zinc oxide depth profiling. It is reasonable to suspect that this H^- signal comes from residual water molecules in the glass. To verify this hypothesis, a thin cover glass (from Ted Pella Inc., $\sim 0.13 \text{ mm}$ thick) was broken into three pieces. Two of them were annealed in vacuum. Then depth profiling measurements were performed. Figure 6 (bottom) shows the results. The original glass shows a high H concentration region at the beginning ($\sim 40 \text{ nm}$ thick), and the bulk H concentration is about 5×10^{19} atoms/ cm^3 , which is lower than the value in bulk Fisher brand glass. After 600°C annealing, the surface water layer is gone, but the bulk H concentration only slightly decreases. After 950°C annealing, the bulk H concentration decreases to about 8×10^{18} atoms/ cm^3 (5.6×10^{18} if the H/D sensitivity difference is considered). This value probably is the detection limit of H in glass samples using our ToF-SIMS instrument for two reasons. First, the annealing temperature of 950°C is close to the liquidus temperature of common glass, and Fig. 6 shows most water

Table 1. Relative sensitivity factors of H and D in Si, ZnO, and common glass using ToF-SIMS

Si			ZnO			Glass		
H	D	Si ^a	H	D	O ^a	H	D	O ^a
1.4 × 10 ²³	1.8 × 10 ²³	5.0 × 10 ²²	7.0 × 10 ²³	8.4 × 10 ²³	4.1 × 10 ²²	NA	1.5 × 10 ²⁴	4.5 × 10 ²²
^a These elements are reference elements in the corresponding matrices. The values are calculated from the densities and chemical formulas of the corresponding materials.								

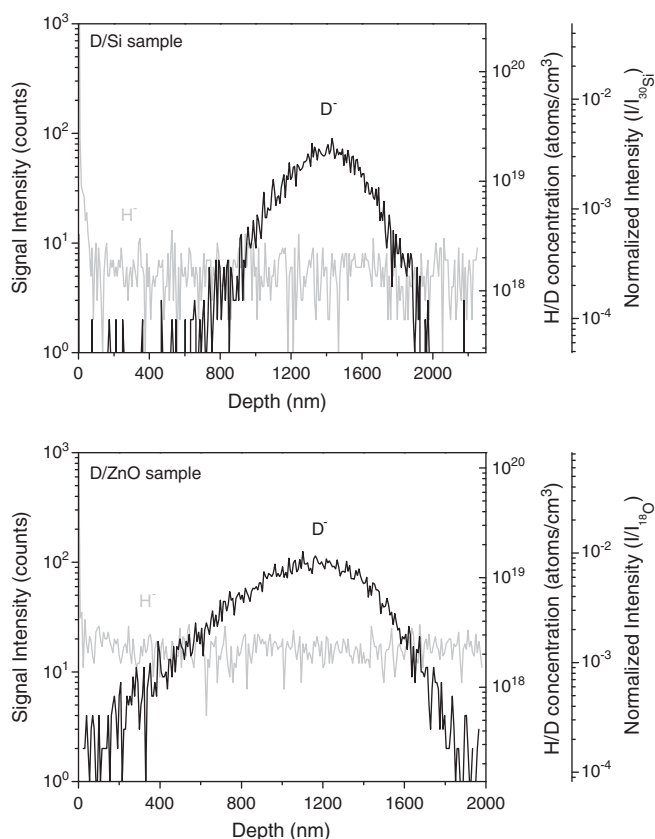


Figure 5. H⁻ and D⁻ depth profiles of a D-implanted silicon sample (top) and a D-implanted zinc oxide sample (bottom). Black line, D⁻ signal; grey line, H⁻ signal. The primary ion beam is a pulsed 25 keV Bi⁺ beam with a 1.0 pA current at a 10 kHz repeating rate, and the collecting time for each data point is 10 s. The pressure in the analysis chamber is $\sim 1.1 \times 10^{-9}$ mbar during sputtering.

molecules have been successfully removed. Second, it should be noted that the pressure in the analysis chamber is about 3.0×10^{-9} mbar during sputtering because charge compensation must be active. Under this vacuum pressure, an H⁻ background of 8×10^{18} atoms/cm³ looks quite reasonable. As a comparison, if charge compensation is active during silicon depth profiling, the pressure in the analysis chamber increases to $\sim 2.5 \times 10^{-9}$ mbar (1.1×10^{-9} without charge compensation), and the H⁻ background signal is $\sim 4 \times 10^{18}$ atoms/cm³ (1.8×10^{18} without charge compensation).

Practically, a widely used strategy to avoid H⁻ background is using D to substitute H. From Figs 5 and 6, it is reasonable to expect 1 order of magnitude improved detection limits if D is used. However, higher detection limits are difficult to achieve because of limited D⁻ signal intensities.

It is well known that magnetic SIMS and quadrupole SIMS provide 1–2 orders of magnitude better sensitivities than ToF-SIMS. However, for hydrogen depth profiling, because detection limits are confined by residual gases, the magnetic and quadrupole SIMS do not show significant advantages as the vacuum systems used on these SIMS instruments are no better than those used on ToF-SIMS instruments. However, for deuterium depth profiling, the magnetic and quadrupole SIMS can regularly reach 1×10^{16} atoms/cm³ because of their strong signal intensities,^[28] at least 1 order of magnitude better than current commercial ToF-SIMS instruments.

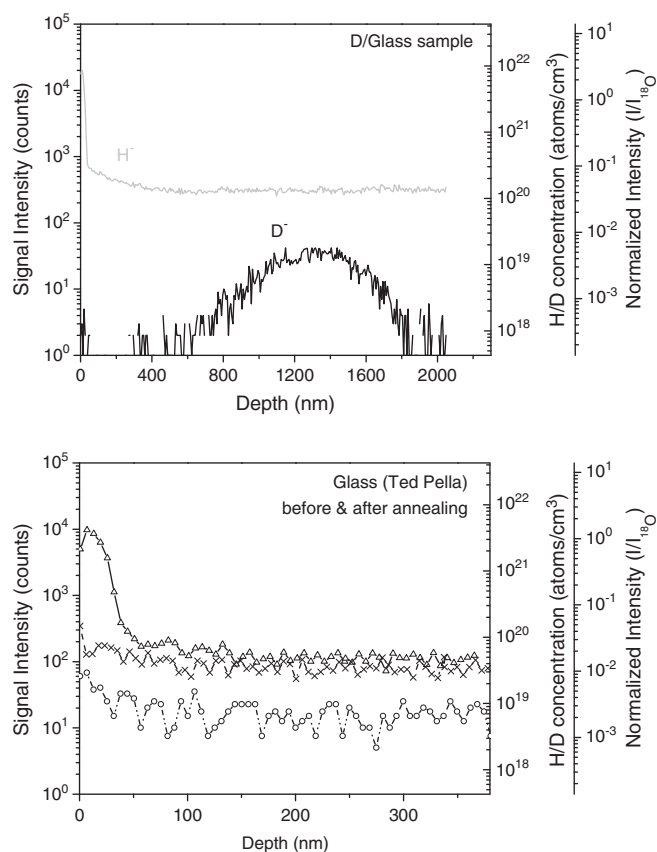


Figure 6. Top, H⁻ and D⁻ depth profiles of a D-implanted Fisher brand glass sample. Black line, D⁻ signal; grey line, H⁻ signal. Bottom, H depth profiles of a Ted Pella thin cover-slip sample before and after annealing. Triangles: before annealing; crosses: after 600 °C, 2 h annealing; circles: after 950 °C, 2 h annealing. The primary ion beam is a pulsed 25 keV Bi⁺ beam with a 1.0 pA current at a 10 kHz repeating rate, and the collecting time for each data point is 10 s. The pressure in the analysis chamber is $\sim 3.0 \times 10^{-9}$ mbar during sputtering.

Summary

In this work, the optimal instrumental settings for hydrogen depth profiling using ToF-SIMS were investigated. To achieve the best detection limits, the following settings are recommended:

- Keep the pressure in the analysis chamber as low as possible.
- Use a Cs⁺ sputtering beam and monitor the H⁻ signal.
- Use the monatomic ion analysis beams with the highest beam currents.
- Use the interlace mode and avoid the noninterlace mode.

In addition, monatomic secondary ions from matrices are recommended as references to normalize the H⁻ signal, and cluster secondary ions are not recommended. The detection limits of H are determined by the pressure of residual gases in the analysis chamber. Our best vacuum in the analysis chamber was about 7×10^{-10} mbar with samples in it, and corresponding detection limits of H in silicon, zinc oxide, and common glass were about 1.3×10^{18} atoms/cm³, 1.8×10^{18} atoms/cm³, and 5.6×10^{18} atoms/cm³, respectively. If using D to substitute H, the detection limits can be 1 order of magnitude better, and they are limited by effective D⁻ signals. In addition, if D is used as a reference to quantify H in a matrix, H may be considered to be about 30% more sensitive than D.

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References

- [1] C. G. Van de Walle, *Phys. Rev. Lett.* **2000**, 85, 1012.
- [2] A. Janotti, C. G. Van de Walle, *Nat. Mater.* **2007**, 6, 44.
- [3] Y. J. Li, T. C. Kaspar, T. C. Droubay, A. G. Joly, P. Nachimuthu, Z. Zhu, V. Shutthanandan, S. A. Chambers, *J. Appl. Phys.* **2008**, 104, 053711.
- [4] Y. J. Li, T. C. Kaspar, T. C. Droubay, Z. Zhu, V. Shutthanandan, P. Nachimuthu, S. A. Chambers, *Appl. Phys. Lett.* **2008**, 92, 152105.
- [5] M. Bruni, D. Bisero, R. Tonini, G. Ottaviani, G. Queirolo, R. Bottini, *Phys. Rev. B* **1994**, 49, 5291.
- [6] R. A. Hakvoort, A. Vanveen, P. E. Mijnders, H. Schut, *Appl. Surf. Sci.* **1995**, 85, 271.
- [7] P. Leveque, P. Pellegrino, A. Hallen, B. G. Svensson, V. Privitera, *Nucl. Instrum. Meth. B* **2001**, 174, 297.
- [8] C. Macchi, S. Mariazzi, G. P. Karwasz, R. S. Brusa, P. Folegati, S. Frabboni, G. Ottaviani, *Phys. Rev. B* **2006**, 74.
- [9] B. C. Bunker, *J. Non-Cryst. Solids* **1994**, 179, 300.
- [10] M. Schreiner, M. Grasserbauer, P. March, *Fresen. Z. Anal. Chem.* **1988**, 331, 428.
- [11] S. Fearn, D. S. McPhail, R. J. H. Morris, M. G. Dowsett, *Appl. Surf. Sci.* **2006**, 252, 7070.
- [12] P. Trocellier, S. Djanarthany, J. Chene, A. Haddi, A. M. Brass, S. Poissonnet, F. Farges, *Nucl. Instrum. Meth. B* **2005**, 240, 337.
- [13] A. Turos, O. Meyer, *Nucl. Instrum. Meth. B* **1984**, 232, 92.
- [14] G. J. Clark, C. W. White, D. D. Allred, B. R. Appleton, C. W. Magee, D. E. Carlson, *Appl. Phys. Lett.* **1977**, 31, 582.
- [15] C. W. Magee, E. M. Botnick, *J. Vac. Sci. Technol.* **1981**, 19, 47.
- [16] J. Bottiger, *J. Nucl. Mater.* **1978**, 78, 161.
- [17] K. Kimura, K. Nakajima, S. Yamanaka, M. Hasegawa, H. Okushi, *Appl. Phys. Lett.* **2001**, 78, 1679.
- [18] R. G. Wilson, *Int. J. Mass Spectrom.* **1995**, 143, 43.
- [19] E. Niehuis, T. Grehl, in *ToF-SIMS: Surface Analysis by Mass Spectrometry*, (Eds: J. C. Vickerman, D. Briggs), IM Publications and Surface-Spectra Limited, Chichester, **2001**, p 753.
- [20] Z. H. Zhu, V. Shutthanandan, Y. J. Li, S. A. Chambers, in *AIP Conf. Proc.*, vol. 1099 (Eds: F. D. McDaniel, B. L. Doyle), **2009**, p. 343.
- [21] G. Boudreault, R. G. Elliman, R. Grotzschel, S. C. Gujrathi, C. Jeynes, W. N. Lennard, E. Rauhala, T. Sajavaara, H. Timmers, Y. Q. Wang, T. D. M. Weijers, *Nucl. Instrum. Meth. B* **2004**, 222, 547.
- [22] D. C. Look, T. C. Droubay, J. S. McCloy, Z. H. Zhu, S. A. Chambers, *J. Vac. Sci. Technol. A* **2011**, 29, 03A102.
- [23] M. L. Yu, *Phys. Rev. Lett.* **1978**, 40, 574.
- [24] F. Kollmer, *Appl. Surf. Sci.* **2004**, 231–2, 153.
- [25] N. Winograd, *Anal. Chem.* **2005**, 77, 142A.
- [26] Z. H. Zhu, V. Shutthanandan, *Surf. Interface Anal.* **2011**, DOI: 10.1002/sia.3776.
- [27] G. Betz, K. Wien, *Int. J. Mass Spectrom.* **1994**, 140, 1.
- [28] K. Ip, M. E. Overberg, Y. W. Heo, D. P. Norton, S. J. Pearton, C. E. Stutz, S. O. Kucheyev, C. Jagadish, J. S. Williams, B. Luo, F. Ren, D. C. Look, J. M. Zavada, *Solid State Electron.* **2003**, 47, 2255.